

Strictly nef divisors on singular varieties, II

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KIAS Geometry Seminar, 25th January 2022

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Today's talk is based on the joint work with Juanyong Wang.

Review previous results

We start from a projective variety $X/\mathbb{C}$.

Definition (Strictly nef)

A $\mathbb{Q}$ -Cartier divisor L on X is said to be **strictly nef**, if $L \cdot C > 0$ for every (integral irreducible) curve C on X .

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Definition (Strictly nef)

A $\mathbb{Q}$ -Cartier divisor L on X is said to be **strictly nef**, if $L \cdot C > 0$ for every (integral irreducible) curve C on X .

Last time, we focused on the case when L is the anti-log canonical divisor $-(K_X + B)$ of a projective klt pair (X, B) and showed the two theorems.

Theorem (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with strictly nef $-(K_X + \Delta)$. Then X is rationally connected. Moreover, the augmented irregularity $q^\circ(X) = 0$.

Theorem (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. If $\dim X = 3$, then $-(K_X + \Delta)$ is ample.

Review previous results

Today, we will study Serrano's conjecture, mainly on the threefold case.

Serrano's conjecture

Let X be a smooth projective variety, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor. Then $K_X + tL_X$ is ample for $t \gg 1$.

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Serrano's conjecture

Let X be a smooth projective variety, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor. Then $K_X + tL_X$ is ample for $t \gg 1$.

Remark (from the viewpoints of cone theory)

Let $N_1(X)$ be the space generated by real 1-cycles of X modulo numerical equivalence. Then the numerical class of every Cartier divisor (up to a multiple) on X is in 1-1 correspondence with the hyperplane of $N_1(X)$ passing through the origin. Applying Kleiman's criterion, a divisor L is ample if and only if the Mori cone $\overline{NE}(X)$ (the cone of effective real 1-cycles) is strictly contained (except for the original) in the half-space $L_{>0}$. If we denote by $H(L)$ the zero locus of the hyperplane defined by L , then Serrano's conjecture is equivalent to $H(L) \cap H(K_X)$ being disjoint with $\overline{NE}(X) \setminus \{0\}$ (the converse direction needs cone theorem).

Review previous results

Theorem (Serrano, 1995)

Serrano's conjecture has a positive answer for Gorenstein surfaces and Abelian varieties.

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When X is K -trivial, Serrano's conjecture predicts that every strictly nef divisor is ample. For the remaining case in $\dim X = 3$, i.e., X is Calabi-Yau and $L_X \cdot c_2 = 0$, Liu and Svaldi had some further progress recently.

Theorem (Liu-Svaldi, 2020)

Let X be a Calabi-Yau threefold with $c_3(X) \neq 0$. Let D be a strictly nef divisor on X with the numerical dimension $\nu(D) \neq 1$. Then D is ample.

Review previous results

Theorem (Liu-Matsumura, 2021)

Let X be a K -trivial fourfold and L a strictly nef divisor on X . If $\kappa(L) \geq 0$, then L is ample. Moreover, if $\nu(L) \neq 2$ on X , then L is ample.

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Now, we ask the singular (pair) version of Serrano's conjecture.

Question (Singular version of Serrano's conjecture)

Let (X, Δ) be a projective klt pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Is $K_X + \Delta + tL_X$ ample for $t \gg 1$?

- The lower bound of t (in terms of the Cartier index of L_X) is another interesting problem.
- klt singularities might be replaced by lc singularities.
- As observed by Liu, the singular version of Serrano's conjecture in $\dim = 3 \Rightarrow$ Campana-Peternell's conjecture in $\dim = 4$.

Review previous results

The singular (log) version of Serrano's conjecture has a positive answer on surfaces by studying the generalized pairs.

Theorem (surface case; Han-Liu, 2020)

Let (X, Δ) be a projective lc surface pair, and L_X a strictly nef Cartier divisor on X . Then $K_X + \Delta + tL_X$ is ample for every real number $t > 3$.

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- It is still widely open in higher dimensions, even in $\dim X = 3$.
- The above theorem (on the surface case) can be further extended to positive characteristics by running a similar strategy.
- Motivated by (Campana-Chen-Peternell, 2008), we give a partial answer to the singular (pure) version of Serrano's conjecture in dimension three. The pair version is still unknown.

Main result, II

The following is our main result. We treat Kodaira dimensions $\kappa(X) > 0$, $\kappa(X) = 0$ and $\kappa(X) = -\infty$ separately.

Theorem (Wang-Z., 2021)

Let X be a normal projective threefold with only klt singularities, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor on X . Suppose that one of the following conditions holds.

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- (1) The Kodaira dimension $\kappa(X) \geq 1$.
- (2) The augmented irregularity $q^\circ(X) > 0$, and either X is $\mathbb{Q}$ -factorial or X has only canonical singularities.

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- (3) The Kodaira dimension $\kappa(X) = 0$, X has only terminal singularities, and $L_X \cdot c_2(X) \neq 0$.

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- (2) The augmented irregularity $q^\circ(X) > 0$, and either X is $\mathbb{Q}$ -factorial or X has only canonical singularities.
- (3) The Kodaira dimension $\kappa(X) = 0$, X has only terminal singularities, and $L_X \cdot c_2(X) \neq 0$.
- (4) X is uniruled with only isolated $\mathbb{Q}$ -factorial Gorenstein canonical singularities (to avoid the small contractions).

Then $K_X + tL_X$ is ample for sufficiently large $t \gg 1$.

Main result, II

Terminal threefolds having isolated singularities, we have:

Corollary (Wang-Z., 2021)

Let X be a $\mathbb{Q}$ -factorial Gorenstein terminal projective threefold, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Then $K_X + tL_X$ is ample for sufficiently large $t \gg 1$ unless X is weak Calabi-Yau and $L_X \cdot c_2(X) = 0$ where $c_2(X)$ is the birational second Chern class on X .

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- The **augmented irregularity** $q^\circ(X)$ is defined as the maximum of the irregularities $q(\tilde{X}) := h^1(\tilde{X}, \mathcal{O}_{\tilde{X}})$, where $\tilde{X} \rightarrow X$ runs over the (finite) quasi-étale covers of X (Nakayama-Zhang, 2010).
- A normal projective klt variety X is said to be **weak Calabi-Yau** if $K_X \sim_{\mathbb{Q}} 0$ and $q^\circ(X) = 0$ (Nakayama-Zhang, 2010).
- The birational second Chern class is defined by its resolution, which is minimal in codimension two (Shepherd-Barron and Wilson, 1994).

Key ideas for the proofs – From effectiveness to ampleness

Lemma 1 ($\mathbb{Q}$ -factorial pure case)

Let X be a $\mathbb{Q}$ -factorial normal projective klt threefold, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor on X . Suppose that there exists some a and b such that $aL_X + bK_X$ is numerically equivalent to a non-zero effective divisor. Then $K_X + tL_X$ is ample for $t \gg 1$.

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Lemma 2 (non- $\mathbb{Q}$ -factorial case)

Let (X, Δ) be a projective klt threefold pair. Let L_X be a strictly nef $\mathbb{Q}$ -divisor on X . Suppose that one of the following holds.

- L_X is numerically equivalent to a non-zero effective divisor.
- $\Delta = 0$, X has only canonical singularities, and $aL_X + bK_X$ is numerically equivalent to a non-zero effective divisor for some a and b .

Then $K_X + \Delta + tL_X$ is ample for sufficiently large t .

We are not able to deal with the general non- $\mathbb{Q}$ -factorial case without assuming any further conditions.

Key ideas for the proofs – The case $\kappa(X) \geq 1$

Theorem (Wang-Z., 2021)

Let (X, Δ) be a projective klt threefold pair, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor on X . Suppose that the log Kodaira dimension $\kappa(X, K_X + \Delta) \geq 1$. Then $K_X + \Delta + tL_X$ is ample for sufficiently large t .

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- $K_X + \Delta + tL_X$ is strictly nef for $t \gg 1$ by applying the cone theorem.
- $K_X + \Delta + tL_X$ is numerically equivalent to an effective divisor by applying a result of Lazić and Peternell.

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As a higher dimensional analogue, we show the following.

Proposition (Wang-Z., 2021)

Let (X, Δ) be a projective canonical pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose the litaka dimension $\kappa(a(K_X + \Delta) + bL_X) \geq \dim X - 2$ for some $a, b \geq 0$. Then $(K_X + \Delta) + tL_X$ is ample for sufficiently large t .

Key ideas for the proofs – The case $q^\circ(X) > 0$

Technical result (Campana-Chen-Peternell, 2008; Wang-Z., 2021)

Let (X, Δ) be a projective klt threefold pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose that $q(X) > 0$. Then $K_X + \Delta + tL_X$ is numerically equivalent to a non-zero effective divisor for some $t \gg 1$.

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Theorem (Wang-Z., 2021)

Let X be a normal projective klt threefold with the augmented irregularity $q^\circ(X) > 0$, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose that either X is $\mathbb{Q}$ -factorial, or X has only canonical singularities. Then $K_X + tL_X$ is ample for sufficiently large t .

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- Pass X to its quasi-étale cover X' such that $q(X') > 0$. By the technical result, $K_{X'} + tL_{X'}$ is num-effective for some $t \gg 1$.
- If X is canonical, then so is X' ; $K_{X'} + tL_{X'}$ is ample by Lemma 2.
- Suppose X is $\mathbb{Q}$ -factorial. Then $K_X + tL_X$ is weakly num-effective. Since X is $\mathbb{Q}$ -factorial, $K_X + tL_X$ is num-effective. Then we are done.

Key ideas for the proofs – The case $\kappa(X) = 0$

Theorem (Wang-Z., 2021)

Let X be a normal projective terminal variety, and L_X a strictly nef divisor on X . If $\kappa(X) = 0$ and $L_X \cdot c_2(X) \neq 0$, then $K_X + tL_X$ is ample for $t \gg 1$.

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- We may assume $q^\circ(X) = 0$ and $K_X \equiv 0$; otherwise, K_X would be num-effective.
- By the abundance, we may further assume $K_X \sim_{\mathbb{Q}} 0$ and thus X is weak Calabi-Yau (Nakayama, 2004) or (Gongyo, 2010).
- On the one hand, after passing X to its “maximally” quasi-étale cover in the sense of Greb-Kebekus-Peternell, we can show that any finite dimensional linear representation of $\pi_1(X_{\text{reg}})$ has abelian image.
- On the other hand, Miyaoka’s pseudo-effectivity of $c_2(X)$ and some further calculation give $h^1(X, -2L_X) \neq 0$. By a theorem of Serrano (with a few modification), there exists a finite dimensional linear representation of $\pi_1(X_{\text{reg}})$ whose image is not abelian.

Key ideas for the proofs – The uniruled case

Now we consider the uniruled case with $q^\circ(X) = 0$. Before that, we first recall the following lemma.

Lemma (Gloria Della Noce, 2014)

Let X be a $\mathbb{Q}$ -factorial Gorenstein normal projective threefold with only isolated canonical singularities. Let $\varphi : X \rightarrow Y$ be a birational K -negative extremal contraction whose fibres are at most 1-dim. Then

- (1) The exceptional $E := \text{Exc}(\varphi)$ is a disjoint union of prime divisors, φ is a composition of divisorial contractions mapping exceptional divisors to curves, and Y has only $\mathbb{Q}$ -factorial isolated canonical singularities.
- (2) There exists a finite subset $T \subseteq Y$ such that is the simultaneous blow-up of smooth curves.
- (3) $K_X = \varphi^*(K_Y) + \sum E_i$ with E_i being exceptional.

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Why we need “Gorenstein canonical with only isolated singularities”?

Key ideas for the proofs – The uniruled case

When running the first step of the MMP, there does not appear small contractions. Now, we shall run the first step of the MMP.

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Fano contraction to curve:

Let (X, Δ) be a projective klt threefold pair, and L_X a strictly nef divisor on X . Suppose $\rho(X) \leq 2$. Then $K_X + \Delta + tL_X$ is ample for $t \gg 1$. In this case, we may assume that $K_X + \Delta$ and L_X are not parallel.

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Fano contraction to surface:

Let X be a $\mathbb{Q}$ -factorial normal projective threefold with only isolated klt singularities, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose there is an elementary K_X -negative contraction $\pi : X \rightarrow S$ onto a normal projective surface S . Then $K_X + tL_X$ is ample for $t \gg 1$.

The main idea is to construct a linear combination $aL_X + bK_X$, which is numerically equivalent to the pullback of an effective divisor from S .

Key ideas for the proofs – The uniruled case

Moreover, we generalize the above theorem to the conic bundle case:

Conic bundle to surface

Let X be a $\mathbb{Q}$ -factorial Gorenstein normal projective threefold with isolated canonical singularities, and L_X a strictly nef divisor on X . Suppose X admits an equi-dimensional K_X -negative extremal contraction $\pi : X \rightarrow S$ onto a normal projective surface such that $K_X + uL_X = \pi^*M$ for some $u \in \mathbb{Q}$ and some $\mathbb{Q}$ -Cartier divisor M on S . Then $K_X + tL_X$ is ample for $t \gg 1$. This is for the usage of the last step.

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Finally, we consider the divisorial contraction case.

Divisorial contraction mapping the exceptional divisor to point

Let (X, Δ) be a $\mathbb{Q}$ -factorial normal projective klt threefold, and L_X a strictly nef divisor. Suppose there is a divisorial contraction $\pi : X \rightarrow X'$ of a $(K_X + \Delta)$ -negative extremal ray, mapping the exceptional divisor to a point. Then $K_X + \Delta + tL_X$ is ample for $t \gg 1$.

Key ideas for the proofs – The uniruled case

Therefore, we are left to show the following theorem.

Divisorial contraction mapping the exceptional divisor to curve

Let X be a $\mathbb{Q}$ -factorial Gorenstein normal projective uniruled threefold with isolated canonical singularities, and L_X a strictly nef divisor. Suppose all the elementary K_X -negative extremal contractions are divisorial, mapping the exceptional divisors to curves. Then $K_X + tL_X$ is ample for $t \gg 1$.

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- We start from a divisorial contraction $\pi : X \rightarrow X_2$ with the exceptional divisor E . By the previous lemma, X_2 is still a $\mathbb{Q}$ -factorial Gorenstein threefold with only isolated canonical singularities.
- If X_2 admits a Fano contraction or a divisorial contraction mapping the exceptional divisor to a point, then by some calculation, we can show that $K_X + tL_X$ is ample.
- If X_2 admits a divisorial contraction mapping the new exceptional divisor E_2 to a curve and $E_2 \cap \pi(E) \neq \emptyset$, then by some calculation, we can show that $K_X + tL_X$ is ample.

Key ideas for the proofs – The uniruled case

$$X = X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow \cdots \rightarrow X_n \rightarrow Y$$

- Inductively, we construct a series of blow-downs $X \rightarrow X_n$ along disjoint curves such that X_n admits a Fano fibration $X_n \rightarrow Y$, noting that X is uniruled. Then we are done by the previous lemma.
- As another technical point, we need to construct a linear combination $D := aK_X + bL_X$ such that every contraction is D -trivial so that it is the pullback of some effective divisor from Y .

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Theorem (Wang-Z., 2021)

Let X be a $\mathbb{Q}$ -factorial Gorenstein normal projective uniruled threefold with isolated canonical singularities, and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor on X . Then $K_X + tL_X$ is ample for sufficiently large $t \gg 1$.

Open Question

Question 1

Let X be a weak Calabi-Yau fourfold (in the sense that $K_X \sim_{\mathbb{Q}} 0$ and $q^{\circ}(X) = 0$), and L_X a strictly nef $\mathbb{Q}$ -Cartier divisor on X . Is L_X ample?

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Compared with the smooth case, which is almost solved by Liu and Matsumura, the remaining case: $\nu(L_X) = 2$ and $L_X^4 = c_2(X) \cdot L_X^2 = 0$ should be challenging. Besides, the generalized Riemann-Roch formula is not so easy according to my consideration.

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Question 2

In the general case when X is only assumed to be klt, can we prove Serrano's conjecture by running the generalized minimal model program? Alternatively, can we prove the $\mathbb{Q}$ -effectiveness of $K_X + tL_X$ for $t \gg 1$ when $\dim X \geq 3$?

Han-Liu and Lazić-Peternell have obtained some inspiring results on this topic without assuming strict nefness.

Thank you for your attention!